

Shubhadeep Mandal

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EDUCATION

International Institute of Information Technology, Hyderabad <i>Telangana, India</i> <i>Master of Technology in Computer Science and Engineering</i>	July 2025 – Present <i>CGPA: 9.12</i>
Indian Institute of Information Technology, Ranchi <i>Ranchi, India</i> <i>Bachelor of Technology (Hons.) in Computer Science and Engineering</i>	Aug 2020 – Jun 2024 <i>CGPA: 9.05</i>

PROJECTS

Peer-to-Peer Distributed File Sharing System <i>C++, Sockets, Multithreading, OpenSSL</i>	🔗
– Built a BitTorrent-style P2P system in C++, achieving an 8.5x speedup on 1GB transfers (193s to 22.8s) across 5 parallel peers. – Engineered out-of-order parallel chunk writing via <code>pwrite()</code> and dynamic bitfield generation for simultaneous partial seeding. – Architected Primary-Backup tracker failover with state synchronization, session flushing to prevent split-brain, and SHA-1 checks.	
FOOL: POSIX-Compliant Interactive Shell <i>C++, POSIX APIs, Systems Programming, Linux</i>	🔗
– Engineered a modular UNIX shell in C++ supporting chained pipelines (<code>pipe</code>, <code>dup2</code>), quote parsing, and I/O redirection. – Implemented job control to manage concurrent foreground and background tasks, preventing terminal deadlocks and zombie accumulation. – Built an interactive terminal line editor with history navigation and non-blocking tab autocompletion for instant command discovery.	
AmbiStory: NLP Plausibility & Disambiguation Engine <i>Python, PyTorch, Transformers, Embeddings</i>	🔗
– Achieved the best non-LLM solution on the AmbiStory problem (SemEval 2026), reaching $\rho = 0.535$ and 72.2% test accuracy. – Engineered a 202-dim vector fusing multi-context sentence embeddings, NLI priors, and PCA interactions into an ordinal regression head. – Mitigated selection bias ($+0.265 \rho$) via homonym-grouped 5-fold CV, annotator distribution KL loss, and Stochastic Weight Averaging (SWA).	
RookDB: In-House Relational Storage Engine <i>Rust, Storage Engine, Buffer Manager, Disk I/O</i>	🔗
– Contributed to IIIT-H's in-house Rust RDBMS, architecting the type system, SQL-99 comparisons, and tuple memory layouts. – Engineered explicit <code>RowLayout</code> structures with bit-packed null bitmaps, accelerating in-memory tuple deserialization. – Integrated nullable row serialization with buffer-pool paging and authored benchmark suites profiling latency against PostgreSQL.	

TECHNICAL SKILLS

Languages: C/C++, Rust, Python, TypeScript, JavaScript, SQL, Bash.
Systems & OS: POSIX APIs, Sockets, Linux/Unix, Multithreading, Git, Storage Engines.
AI, ML & NLP: Transformers (DeBERTa, MPNet), PyTorch, Hugging Face, NLI, Sentence-Transformers, Scikit-learn.
Web Technologies: Next.js, React, Node.js, Convex, Prisma, PostgreSQL, WebSockets.

ACHIEVEMENTS

- **B.Tech Rank Holder:** Ranked 2nd in IIIT Ranchi by GPA in B.Tech CSE.
- **PGEE 2025:** Secured All-India Rank 100 in IIIT-Hyderabad Post Graduate Entrance Examination (PGEE).
- **Megathon 2025:** Third rank team in Megathon 2025 in the mobile computing track.

INTERESTS

- Web Novel and Manga Enthusiast.
- Chess and Badminton.